Chapter 7 Notes

Section 1

In the Remarks are page 91, it is said that there is an "easy induction" showing that R_{n+1} has 2^n elements. However, this is clearly not the case. For, as remakred, R_3 has 4 elements. Then R_4 has $2^4 = 16$ elements, which is not equal to $2^3 = 8$.

The proof of Proposition 1.3 is problematic. It uses Cantor's Theorem, though this hasn't even been stated (let alone proven) in the context of Zermelo-Frankel Universes. Proposition 1.3 can also be proven without using Cantor's Theorem. In particular, we first show that $R_\alpha \notin R_\alpha$.

Proof. By transfinite induction. Clearly this holds for 0. Suppose it holds for some α. Then, if $R_{\alpha+1} \in R_{\alpha+1}$, by definition of $R_{\alpha+1}$, $R_{\alpha+1}$ would be a subset of R_α. But $R_\alpha \in R_{\alpha+1}$. Thus this would mean that $R_\alpha \in R_\alpha$, contradicting the induction assumption. Hence $R_{\alpha+1} \notin R_{\alpha+1}$. On the other hand, suppose it held for all ordinals less than some limit ordinal λ. Then, by definition, $R_\lambda = \cup_{\alpha<\lambda} R_\alpha$. Suppose $R_\lambda \in R_\lambda$. Then, for some $\alpha < \lambda$, $R_\lambda \in R_\alpha$. But $R_{\alpha+1} \subset R_\lambda$, and $R_\alpha \in R_{\alpha+1}$, meaning $R_\alpha \in R_\lambda$. Thus, because R_α is transitive, $R_\alpha \in R_\alpha$, contradicting the induction assumption. Thus $R_\lambda \notin R_\lambda$, completing the transfinite induction. □

Then, in the proof of Proposition 1.3, $R_\alpha \neq R_{\alpha+1}$ follows for the following reason: We know that $R_\alpha \in R_{\alpha+1}$ and $R_\alpha \notin R_\alpha$ (by the above result), meaning $R_\alpha \neq R_{\alpha+1}$.

Section 2

Note that, at a minimum, the R_α sequence discussed in this section is an On-sequence of subsets of V (and of course, all On-sequences for any particular class are also On-sequences for V itself).

Note that the proof of O_6 uses Theorem 1.18 of Chapter 5.

Section 3

Notes

Here is a proof of part (3) of Q_1: Suppose that both $x \in R_\Omega$ and $y \in R_\Omega$. Let α_1 and α_2 be the ranks of x and y, respectively. Let α be the maximum of α_1, α_2. Thus $x \in R_{\alpha^+}$ and $y \in R_{\alpha^+}$. Hence $\{x, y\} \in R_{\alpha^{++}}$, meaning $\{x, y\}$ has rank, hence $\{x, y\} \in R_\Omega$.

Here is a proof of part (5) of Q_2: Suppose that $x \in R_\Omega$ (say it is of rank α). Then $x \in R_{\alpha^+}$, meaning, by Q_2 (3), $\cup x \in R_\alpha$, and thus $\cup x$ has rank.

Here are the proofs of part (2) and (3) of Q_3:

(2) Suppose $x \in R_\lambda$. Then $x \in R_\alpha$ for some $\alpha < \lambda$. Hence $x \in R_\alpha^+$, meaning $\mathcal{P}(x) \in R_{\alpha+2}$ by Q_3 (1), and because $\alpha + 2 < \lambda$, $\mathcal{P}(x) \in R_\lambda$.

(3) Suppose $x \in R_\Omega$. Then $x \in R_{\alpha^+}$ for some α, meaning, by Q_3 (1), $\mathcal{P}(x) \in R_{\alpha+2}$, and thus $\mathcal{P}(x)$ has rank.

Section 4

Notes

Here is why if C is a class of ordinals such that there is no ordinal greater than all ordinals in C, then C cannot be a set. For suppose it were a set. Then C would be a chain of ordinals, hence $\cup C$ would be an ordinal. But $\cup C = On$. Obviously $\cup C \subseteq On$. On the other hand, for any ordinal α, there must be some ordinal $\beta \in C$ such that $\alpha < \beta$, meaning $\alpha \in \cup C$. Thus this would mean that On was an ordinal, which is impossible.

Note that if x has rank and $y \subseteq x$, then y also has rank. Say x has rank α. Then $x \subseteq R_\alpha$, meaning all elements of x are elements of R_α, hence have rank. Now, x is a set, so y is also a set. Thus y is a set of elements which have rank, meaning, by Q_4, y has rank.

Here is why $\omega \times 2$ is the least limit ordinal $> \omega$. For suppose we had some limit ordinal λ such that $\omega < \lambda < \omega \times 2$. Thus, for some $x \in \omega \times 2$, $\lambda \in \omega \times 2$. By definition of $\omega \times 2$, either λ is a natural number, or $\lambda = \omega + n$ for some natural number n. The former case is ruled out because λ is a limit ordinal. Because $\lambda \neq \omega$ by assumption, this would mean that $\lambda = \omega + n$ for some $n > 0$. But this would mean that λ was a successor ordinal, a contradiction. Hence there can be no limit ordinal λ such that $\omega < \lambda < \omega \times 2$.

In the proof of Theorem 4.1, here is why $\omega \in R_\lambda$ (if $\lambda > \omega$ and λ is a limit ordinal). This is because $R_{\omega^+} \subset R_\lambda$, and $\omega \in R_{\omega^+}$.

In the proof of Theorem 4.2, here is why, if Z is a Zermelo universe and if $\beta \in Z$, then $\beta^+ \in Z$. For suppose $\beta \in Z$. Then, because Z is a Zermelo universe, $\{\beta\} \in Z$, and also $\{\beta, \{\beta\}\} \in Z$, and thus also $\cup\{\beta, \{\beta\}\} \in Z$. But $\cup\{\beta, \{\beta\}\} = \beta^+$, meaning $\beta^+ \in Z$.

It can also be proven that R_Ω does not have rank.

Proof. For suppose it had some rank α. Thus $R_\Omega \subseteq R_\alpha \subset R_{\alpha^+}$, meaning there is some element x in R_{α^+} not in R_Ω. But, because $x \in R_{\alpha^+}$, x has rank (by P_4), meaning, by the definition of R_Ω, $x \in R_\Omega$, a contradiction. Thus it cannot be that R_Ω has rank. □

With this, it easily follows that R_Ω is not a set. For suppose it were. Then it would be a set of elements with rank, meaning, by O_6, R_Ω would have rank, which is impossible. Thus R_Ω is not a set.

Suppose we tried to define a new On-sequence as follows: (1) $\mathcal{O}_0 = R_\Omega$;
(2) $\mathcal{O}_{\alpha^+} = \mathcal{P}(\mathcal{O}_\alpha)$;
(3) $\mathcal{O}_\lambda = \cup_{\alpha<\lambda} R_\alpha$ (for any limit ordinal λ).
This doesn't work, because R_Ω is not a set.

We can also prove that none of the R_α's are Zermelo-*Fraenkel* universes.

Proof. For suppose that some R_α was a Zermelo-Fraenkel universe. Then, for any function F from sets in R_α to sets in R_α, and for any $x \in R_\alpha$, $F''(x)$ is a set in R_α. But consider the function identity function f from R_α onto R_α. Then, because R_α is a set, $f(R_\alpha)''$ is in R_α. But, by definition of f, $f(R_\alpha)''$ is just R_α. Thus we would have $R_\alpha \in R_\alpha$, which contradicts the fact that R_α has rank α. Hence R_α is not a Zermelo-Fraenkel universe. □

This is a question I could not solve: Is it true that no Zermelo-Fraenkel universe can be a set?

Certinaly, we can show that for a Zermelo-Fraenkel universe U, $U \notin U$.

Proof. For suppose $U \in U$. Then $\mathcal{P}(U) \in U$, meaning $\mathcal{P}(U) \subseteq U$, hence $\mathcal{P}(U)$ could be put into a 1-1 correspondence with a subset of U. But, by Cantor's Theorem, this is impossible. Hence $U \notin U$. □

I was also able to prove the result that for any Zermelo-Fraenkel universe U that contains all ordinals of V, U is not a set.

Proof. For suppose U was a set. By Exercise 4.1, it follows that $R_\Omega \subseteq U$, which would mean that R_Ω is a set, a contradiction. Hence U is not a set. □

Exercise 4.1

We show that R_Ω is the intersection of all subclasses C of V such that C is a Zermelo-Fraenkel universe containing all ordinals of V.

(1) On the one hand, suppose that $x \in R_\Omega$. We need to show that x is in the intersection of all subclasses C of V such that C is a Zermelo-Fraenkel universe containing all ordinals of V. So consider any sublcass C of V such that C is a Zermelo-Fraenkel universe containing all ordinals of V.

We first show that $R_\alpha \in C$ for all α. Becuase $R_0 = \emptyset$, and $\emptyset \in C$, $R_0 \in C$. Suppose now for some α, $R_\alpha \in C$. Then $R_{\alpha^+} = \mathcal{P}(R_\alpha)$. But, by the power axiom, $\mathcal{P}(R_\alpha) \in C$, meaning $R_{\alpha^+} \in C$. On the other hand, suppose that $R_\alpha \in C$ for all $\alpha < \lambda$ for a limit ordinal λ. Then $R_\lambda = \cup_{\alpha<\lambda} R_\alpha$. Now, $\lambda \in C$ by assumption, and $\cup_{\alpha<\lambda} R_\alpha$ can be put into a 1-1 correspondence with λ, hence is in C (by the axiom of substitution). Thus $R_\lambda \in C$. Hence, for all ordinals α, $R_\alpha \in C$.

Having shown this, because $x \in R_\Omega$, $x \in R_{\alpha+1}$ for some α. But $R_{\alpha+1} \in C$, and because C is supercomplete, $R_{\alpha+1} \subset C$, meaning $x \in C$. Thus we have shown that R_Ω is a subset of the intersection of all subclasses C of V such that C is a Zermel-Fraenkel universe containing all ordinals of V.

On the other hand, suppose that x is in the intersection of all subclasses C of V such that C is a Zermelo-Fraenkel universe containing all ordinals of V. Well, R_Ω is a subclass of V that is a Zermelo-Fraeknel universe containing all ordinals (by P_7 and Theorem 4.1), meaning $x \in R_\Omega$. Hence the intersection of all subclasses C of V such that C is a Zermelo-Fraeknel universe containing all ordinals of V is a subclass of R_Ω.

Consequently, R_Ω is the intersection of all subclasses C of V such that C is a Zermelo-Fraenkel universe containing all ordinals of V.

Exercise 4.2

I did not know what a Cauchy sequence was. Instead of solving the exact problem, I proved that the set of all rationals is a set with rank.

First, every fraction is a set with rank. For consider some fraction $\langle m, n\rangle$. Then m and n are both ordinals. Without loss of generality, suppose that $m \geq n$. Thus $m \in R_{m+1}$ and $n \in R_{m+1}$. Now, $\langle m, n\rangle = \{\{m\}, \{m, n\}\}$. Thus $\langle m, n\rangle \in \mathcal{P}(\mathcal{P}(R_{m+1}))$, that is, $\langle m, n\rangle \in R_{m+3}$, hence $\langle m, n\rangle$ has rank.

Now consider $\mathbb{F}$. Clearly $\mathbb{F}$ does not have rank n for a natural number n. We show that $\mathbb{F}$ has rank ω. That is, we show that $\langle m, n\rangle \in R_\omega$ for all positive integers m and n, which would mean that $\mathbb{F} \subseteq R_\omega$.

Consider any $\langle m, n\rangle$, that is, $\{\{m\}, \{m, n\}\}$. Without loss of generality, let m be the maximum of m, n. Then $m \in R_{m+1}$ and $n \in R_{m+1}$. Hence $\{\{m\}, \{m, n\}\} \in R_{m+3}$, meaning (by the definition of R_ω) that $\langle m, n\rangle \in R_\omega$. Thus $\mathbb{F}$ has rank ω.

For our purposes, let $\mathbb{Q}$ be the set of all sets of fractions, that is, $\mathcal{P}(\mathbb{F})$. We show that $\mathbb{Q}$ has rank $\omega + 1$. Clearly $\mathbb{Q} \subseteq R_{\omega+1}$ (because $R_{\omega+1} = \mathcal{P}(R_\omega)$ and $\mathbb{F} \subseteq R_\omega$, and thus $\mathcal{P}(\mathbb{F}) \subseteq \mathcal{P}(R_\omega)$). Clearly $\mathbb{Q}$ does not have rank n where n is a natural number. Then the only possiblity other than $\omega + 1$ would be ω. But by P_6, this would mean (because $\mathbb{F} \in \mathbb{Q}$) that

$\mathbb{F}$ had rank lower than ω, contradicting the fact that $\mathbb{F}$ has rank ω. Hence $\mathbb{Q}$ does not have rank ω, and consequently, $\mathbb{Q}$ has rank $\omega + 1$.

Exercise 4.3

I did not do this, as I did not know what a Cauchy sequence was.